LaTeX Cheat Sheet

Document classes

article	Best suited for exercise sheets.

Used at the very beginning of a document : \documentclass{*class*}. Use \begin{document} to start contents and \end{document} to end the document.

Common documentclass options

10pt/11pt/12pt	Font size.
letterpaper/a4paper	Paper size.
twocolumn	Use two columns.
twoside	Set margins for two-sided.
landscape	Landscape orientation.
draft	Double-space lines.

Usage : \documentclass[*opt,opt*]{*class*}.

Recommended packages

N.B. In the digital version of this cheat sheet all package names link to their respective CTAN page for quick access to the full documentation.

babel	Support for non-English languages. See below for French typesetting.
geometry	Page layout (margins, etc.)
multicol	Use n columns : \begin{multicols}{n}.
fancyhdr	Customize header and footer
mathtools	Advanced math typesetting. Includes amsmath.
graphicx	Show image : \includegraphics[width=*x*]{*file*}.
hyperref	Insert URL : \url{*https://...*} or hyperlink : \href{URL}{text}.
enumitem	Customize enumerate.
tabularx	Stretchable X columns in tables.
tabulary	Automatically balanced column widths.
tabto	Tab stops (compatible with enumerations).
tikz	**T**ikZ **i**st **k**ein **Z**eichenprogramm. Make beautiful and precise scientific drawings. Full online documentation : https://tikz.dev
tkz-euclide	Tools for drawing Euclidean geometry; based on TikZ/PGF.
pgfplots	Create various high-quality plots; based on TikZ/PGF.
beamer	For creating slides.

Use before \begin{document}.
Usage : \usepackage{*package*}

Title

\author{*text*}	Author of document.
\title{*text*}	Title of document.
\date{*text*}	Date.

These commands go before \begin{document}. The declaration \maketitle goes at the top of the document.

Miscellaneous

\pagestyle{empty}	Empty header, footer and no page numbers.
\tableofcontents	Add a table of contents here.

Document structure

\part{*title*}	\subsubsection{*title*}
\chapter{*title*}	\paragraph{*title*}
\section{*title*}	\subparagraph{*title*}
\subsection{*title*}	

Use a *, as in \section*{*title*}, to not number a particular item—these items will also not appear in the table of contents.

Text environments

\begin{comment}	Comment (not printed). Requires verbatim package.
\begin{quote}	Indented quotation block.
\begin{quotation}	Like quote with indented paragraphs.
\begin{verse}	Quotation block for verse.

Lists

\begin{enumerate}	Numbered list.
\begin{itemize}	Bulleted list.
\begin{description}	Description list.
\item *text*	Add an item.
\item[*x*] *text*	Use *x* instead of normal bullet or number. Required for descriptions.

References

\label{*marker*}	Set a marker for cross-reference, often of the form \label{sec:item}.
\ref{*marker*}	Give section/body number of marker.
\pageref{*marker*}	Give page number of marker.
\footnote{*text*}	Print footnote at bottom of page.

Floating bodies

\begin{table}[*place*]	Add numbered table.
\begin{figure}[*place*]	Add numbered figure.
\begin{equation}[*place*]	Add numbered equation.
\caption{*text*}	Caption for the body.

The *place* is a list valid placements for the body. t=top, h=here, b=bottom, p=separate page, !=place even if ugly. Captions and label markers should be within the environment.

Text properties

Font face

Command	*Declaration*	*Effect*
\textrm{*text*}	{\rmfamily *text*}	Roman family
\textsf{*text*}	{\sffamily *text*}	Sans serif family
\texttt{*text*}	{\ttfamily *text*}	Typewriter family
\textmd{*text*}	{\mdseries *text*}	Medium series
\textbf{*text*}	{\bfseries *text*}	**Bold series**
\textup{*text*}	{\upshape *text*}	Upright shape
\textit{*text*}	{\itshape *text*}	*Italic shape*
\textsl{*text*}	{\slshape *text*}	*Slanted shape*
\textsc{*text*}	{\scshape *text*}	SMALL CAPS SHAPE
\emph{*text*}	{\em *text*}	*Emphasized*
\textnormal{*text*}	{\normalfont *text*}	Document font
\underline{*text*}		Underline

The command (t*ttt*) form handles spacing better than the declaration (t*ttt*) form.

Font size

\tiny	tiny	\Large	Large
\scriptsize	scriptsize	\LARGE	LARGE
\footnotesize	footnotesize	\huge	huge
\small	small	\Huge	Huge
\normalsize	normalsize		
\large	large		

These are declarations and should be used in the form {\small ...}, or without braces to affect the entire document.

Verbatim text

\begin{verbatim}	Verbatim environment.
\begin{verbatim*}	Spaces are shown as ␣.
\verb!text!	Text between the delimiting characters (in this case '!') is verbatim.

Justification

Environment	*Declaration*
\begin{center}	\centering
\begin{flushleft}	\raggedright
\begin{flushright}	\raggedleft

Better typesetting (by enabling hyphenation) can be achieved using the package ragged2e and its environments RaggedRight, RaggedLeft, and Center.

Miscellaneous

\linespread{x} changes the line spacing by the multiplier x.

French typography

Use babel-french :

```
\usepackage[french]{babel}
% fix spacing when using « » in source
\frenchsetup{og=«, fg=»}
% fix spacing in e.g. 1,23
\usepackage{icomma}
```

1^{er}	1\ier	1^{ers}	1\iers	1^{re}	1\iere	1^{res}	1\ieres
2^{e}	2\ieme	2^{es}	2\iemes	M^{me}	M\up{me}	D^{r}	D\up{r}
œ	\oe	Œ	\OE	æ	\ae	Æ	\AE
n^{o}	\no	n^{os}	\nos	N^{o}	\No	N^{os}	\Nos
«	\og{}	»	\fg{}	« Oh ! »	\frquote{Oh!}		

Polyglossia is a modern replacement for babel, but it's only compatible with XeLaTeX and LuaLaTeX.

Text-mode symbols

Symbols

&	\&	_	_	…	\ldots	•	\textbullet
$	\$	^	\^{}	|	\textbar	\	\textbackslash
%	\%	~	\~{}	#	\#	§	\S

Delimiters

' `	" ``	{ \{	[[	((	< \textless
' '	" ''	} \}	]]	))	> \textgreater

Dashes

Name	*Source*	*Example*	*Usage*
hyphen	`-`	X-ray	In words.
en-dash	`--`	1–5	Between numbers.
em-dash	`---`	Yes—or no?	Punctuation.

Line and page breaks

`\\`	Begin new line without new paragraph.
`\\*`	Prohibit pagebreak after linebreak.
`\kill`	Don't print current line.
`\pagebreak`	Start new page.
`\columnbreak`	Start new column.
`\noindent`	Do not indent current line.

Spaces and rules

`~`	Fixed space, disallow linebreak (`N.~Biever`).
`\smallskip`	Small vertical space.
`\medskip`	Medium vertical space.
`\bigskip`	Big vertical space.
`\hspace{`l`}`	Horizontal space of length l (Ex: $l = 20$`pt`).
`\vspace{`l`}`	Vertical space of length l.
`\rule{`w`}{`h`}`	Line of width w and height h.

Rules of width/height 0 are sometimes useful for solving spacing or alignment issues.

Tables and tab stops

Tab stops: `tabbing` vs. `tabto`

`\=` Set tab stop. `\>` Go to tab stop.

Tab stops can be set on "invisible" lines with `\kill` at the end of the line. Normally `\\` is used to separate lines.

The `tabbing` environment is incompatible with `enumerate`/`itemize`. In this case it's better to use the package `tabto` instead:

```
\TabPositions{2cm, 5cm}
Some\tab sample\tab text.
```

`tabular` environment

`\begin{array}[`*pos*`]{`*cols*`}`
`\begin{tabular}[`*pos*`]{`*cols*`}`
`\begin{tabular*}{`*width*`}[`*pos*`]{`*cols*`}`

`tabular` column specification

`l` Left-justified column.
`c` Centered column.
`r` Right-justified column.
`p{`*width*`}` Same as `\parbox[t]{`*width*`}`.
`@{`*decl*`}` Insert *decl* instead of inter-column space.
`|` Inserts a vertical line between columns.

`tabular` elements

`\hline`	Horizontal line between rows.
`\cline{`x-y`}`	Horizontal line across columns x through y.
`\multicolumn{`*n*`}{`*cols*`}{`*text*`}`	A cell that spans *n* columns, with *cols* column specification.

Math mode

For inline math, use `\(...\)` or `$...$`. For displayed math, use `\[...\]` or `\begin{equation}`.

2^x	`2^{x}`	x_0	`x_{0}`
$\frac{x}{y}$	`\frac{x}{y}`	$\sqrt[n]{x}$	`\sqrt[n]{x}`
$\sum_{k=1}^n$	`\sum_{k=1}^n`	$\prod_{k=1}^n$	`\prod_{k=1}^n`
$\int_a^b f$	`\int_a^b f`	$\iint_S$	`\iint_S`
$\vec{u}$	`\vec{u}`	$\overrightarrow{AB}$	`\overrightarrow{AB}`

Matrices and determinants:

```
\begin{pmatrix}
1&2\\   3&4
\end{pmatrix}
```

$$\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$$

```
\begin{vmatrix}
1&2\\   3&4
\end{vmatrix}
```

$$\begin{vmatrix}1 & 2\\ 3 & 4\end{vmatrix}$$

Math-mode symbols

$\leq$	`\leq`	$\geq$	`\geq`	$\neq$	`\neq`	$\approx$	`\approx`
$\times$	`\times`	$\div$	`\div`	$\pm$	`\pm`	$\cdot$	`\cdot`
$^{\circ}$	`^{\circ}`	$\circ$	`\circ`	$\prime$	`\prime`	$\cdots$	`\cdots`
∞	`\infty`	$\neg$	`\neg`	$\wedge$	`\wedge`	$\vee$	`\vee`
$\supset$	`\supset`	$\forall$	`\forall`	$\in$	`\in`	$\rightarrow$	`\rightarrow`
$\subset$	`\subset`	$\exists$	`\exists`	$\notin$	`\notin`	$\Rightarrow$	`\Rightarrow`
$\cup$	`\cup`	$\cap$	`\cap`	$\mid$	`\mid`	$\Leftrightarrow$	`\Leftrightarrow`
$\dot a$	`\dot a`	$\hat a$	`\hat a`	$\bar a$	`\bar a`	$\tilde a$	`\tilde a`
α	`\alpha`	β	`\beta`	γ	`\gamma`	δ	`\delta`
ϵ	`\epsilon`	ζ	`\zeta`	η	`\eta`	ε	`\varepsilon`
θ	`\theta`	ι	`\iota`	κ	`\kappa`	ϑ	`\vartheta`
λ	`\lambda`	μ	`\mu`	ν	`\nu`	ξ	`\xi`
π	`\pi`	ρ	`\rho`	σ	`\sigma`	τ	`\tau`
υ	`\upsilon`	ϕ	`\phi`	χ	`\chi`	ψ	`\psi`
ω	`\omega`	Γ	`\Gamma`	Δ	`\Delta`	Θ	`\Theta`
Λ	`\Lambda`	Ξ	`\Xi`	Π	`\Pi`	Σ	`\Sigma`
Υ	`\Upsilon`	Φ	`\Phi`	Ψ	`\Psi`	Ω	`\Omega`

Sample LaTeX document

```
\documentclass[11pt]{article}
\title{Template}
\author{Name}
\begin{document}
\maketitle

\section{section}
\subsection*{subsection without number}
text \textbf{bold text} text. Some math: $2+2=5$
\subsection{subsection}
text \emph{emphasized text} text. Display math:
\[
\int_0^1 x^2 \,\mathrm{d}x = \frac{1}{3}
\]

A centered table:
\begin{center}
\begin{tabular}{|l|c|r|}
\hline
first  &  row  &  data \\
second &  row  &  data \\
\hline
\end{tabular}
\end{center}
\end{document}
```


https://wch.github.io/latexsheet/
https://gschintgen.github.io/latexsheet/